Spatio-Temporal Analysis of Soccer Match Events Across European Leagues

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***Abstract*—This paper presents a comprehensive spatio-temporal machine learning and data mining analysis of a large-scale soccer match event dataset spanning five top-tier European leagues (England, Spain, Italy, Germany, France) and two international tournaments (FIFA World Cup 2018, UEFA Euro 2016), sourced from the publicly available Wyscout dataset on Kaggle [1]. Building on the spatial analysis in Assignment 1 and the temporal analysis in Assignment 2, this work integrates both dimensions into a unified analytical framework. Spatial features encode pitch-zone statistics, KNN spatial lag of shot counts, and LISA hotspot labels derived from a 10×10 pitch grid. Temporal features capture per-match shot volumes, lag and rolling statistics, half-ratio, and within-match shot location aggregates. Tuned machine learning models (Random Forest, KNN, SVM, Ridge) achieve high performance on spatial regression (SVR $R^2 = 0.951$, Ridge $R^2 = 0.977$) and competitive temporal classification results. Auto-ARIMA with AIC-based stepwise order selection provides data-driven ARIMA parameters per league. Spatial data mining via K-Means ($K = 6$), DBSCAN, and association rule co-location confirms shot clustering near the penalty area. Temporal data mining via PrefixSpan sequential pattern mining, Apriori, and Markov chain shot-state analysis establishes that Pass→Pass is the dominant pre-shot sequence across all competitions. A joint spatio-temporal ablation study, Zone×Time-bin density matrix, seasonal LISA evolution, and PCA on combined features collectively confirm that attacking-third shot hotspots are structurally stable across season thirds and that shot pressure peaks in the 75–90 minute window. The source code is available at [2].**

***Index Terms*—Spatio-Temporal Analytics, Soccer Events, LISA, PrefixSpan, ARIMA, Random Forest, Feature Engineering, Joint Inference, Ablation Study**

I. Dataset Description

A. Overview

The Soccer Match Event Dataset [1] is a publicly available collection on Kaggle containing match event records from five top-tier European football leagues and two international tournaments, originating from Wyscout event feeds. The England dataset contains 643,090 event records; remaining leagues range from 519,357 (Germany) to 647,300 (Italy). The FIFA World Cup 2018 covers 101,756 events across 64 matches and UEFA Euro 2016 covers 78,125 events across 51 matches. Table I presents the full summary.

B. Dataset Structure

Each event record encodes the action type (`eventName`), sub-type (`subEventName`), match and team identifiers (`matchId`, `teamId`), match period, event time in seconds (`eventSec`), and spatial coordinates normalised to a $[0, 100] \times [0, 100]$ pitch frame (`pos_orig_x`, `pos_orig_y`). Temporal context is recovered by joining each event's `matchId` against the corresponding matches CSV, which provides `dateutc` and `gameweek`. This join is the critical preprocessing step that transforms an unordered event table into a time-ordered series. Two temporal granularities are used: gameweek series (≈ 38 points per league season) and match-level series (64 points for World Cup, 51 for Euro 2016).

TABLE I
Dataset Summary Statistics

Dataset	Matches	Total Events	Shots
England	380	643,090	8,450
Spain	380	628,550	7,977
Italy	380	647,300	8,806
Germany	306	519,357	6,896
France	380	632,737	8,326
World Cup	64	101,756	∼1,750
Euro 2016	51	78,125	∼1,400

C. Why This Dataset Is Suitable for Spatio-Temporal Analysis

The dataset is uniquely suited to joint spatio-temporal analysis for five reasons. First, coordinate normalisation to $[0, 100]^2$ ensures a common spatial reference frame across all matches and leagues. Second, `eventSec` provides continuous intra-match temporal resolution enabling analysis at multiple granularities. Third, the multi-competition structure supports comparative analysis across qualitatively different tactical environments: a 38-point regular season versus a short elimination tournament. Fourth, multiple event types (shots, passes, duels, fouls, free kicks) enable stratified spatial analysis and multi-feature temporal regression. Fifth, from the spatial analytics perspective the dataset constitutes a marked spatio-temporal point process, making it ideal for the combined methods described in this paper.

II. Feature Engineering

A. Spatial Feature Engineering

The pitch is discretised into a regular 10×10 grid (100 cells). Each cell forms a spatial observation unit. The spatial feature set `SPATIAL_FEATURES` comprises:

Geometric features: `x_coord`, `y_coord` (cell centroid), `dist_to_goal` (Euclidean distance to attacking goal at $(100, 50)$), and `is_central` (binary flag for $30 \leq y \leq 70$).

Event count and rate features: `pass_count`, `duel_count`, `foul_count`, `shot_rate` (proportion of events that are shots).

Spatial autocorrelation features: `spatial_lag_shots`, the row-standardised KNN ($k = 4$) spatial lag of shot count encoding local neighbourhood intensity. `is_hotspot`, a binary significant High-High ($p < 0.05$) Local Moran's I indicator. Per-league LISA computation yields the hotspot counts: England (1), Spain (0), Italy (1), Germany (0), France (0), World Cup (13), Euro 2016 (0), totalling 700 spatial training rows across all datasets.

B. Temporal Feature Engineering

Temporal features are constructed at the match level for each league. The response variables are `next_shots` for regression and `result_label` (Home Win / Draw / Away Win) for classification.

Event count features: Per-match aggregates of shots, passes, duels, fouls, and free kicks joined from events on `matchId`. Per-league match table sizes after lag dropping: England (379), Spain (379), Italy (379), Germany (305), France (379), World Cup (63), Euro 2016 (50).

Lag and rolling features: Shot and goal lags at $l \in \{1, 2, 3\}$ (`shots_lag1`, `goals_lag1`, etc.) capture recent match history. Rolling mean windows of 3 and 5 (`shots_roll3`, `shots_roll5`) smooth the short-term trend.

Intra-match temporal features: `half_ratio` is the ratio of first- to second-half event counts. `shot_spread_x`, `shot_spread_y` are the standard deviations of shot coordinates within a match. `shot_avg_x` is the mean shot x-coordinate, encoding attacking depth. The last two features bridge the spatial and temporal domains and are used in the joint inference experiments of Section VII.

III. Machine Learning Models

A. Spatial Machine Learning

1) Shot Density Regression : Four regression models predict `shot_count` per grid zone on the England dataset with an 80/20 hold-out split, all tuned via `GridSearchCV` (3-fold CV, MAE scoring). Best hyperparameters: RF (`n_estimators=100, max_depth=None, min_samples_split=2`); KNN (`k=7, weights=distance, metric=manhattan`); Ridge ($\alpha = 1.0$); SVR (`C=10, kernel=linear, gamma=scale`).

Table II presents the results. Ridge achieves the highest $R^2 = 0.977$ while SVR achieves the lowest MAE of 5.73. Figure 1 shows the feature importance and prediction plots.

2) Hotspot Classification: Binary classification predicts whether a grid zone is a significant LISA HH hotspot. Class imbalance is severe (498 non-hotspot vs. 2 hotspot). All four classifiers achieve perfect accuracy and F1 = 1.0 with `class_weight='balanced'`, indicating that the spatial feature signature of hotspot zones is unambiguous. Table III summarises the results.

3) League Multi-class Classification : A 5-class classifier identifies the league of origin from spatial zone features alone. The tuned Random Forest achieves overall accuracy of 0.480. Table IV shows per-league precision, recall and F1. England is the most spatially distinctive league (F1 = 0.67), while Spain and France are harder to separate (F1 = 0.37–0.39), consistent with their similar spatial shot clustering profiles.

TABLE II
SPATIAL REGRESSION RESULTS – ENGLAND (TUNED, 80/20 SPLIT)

Model	MAE	R^2
RF (tuned)	7.78	0.923
KNN (tuned)	14.21	0.731
Ridge (tuned)	6.00	0.977
SVR (tuned)	5.73	0.951

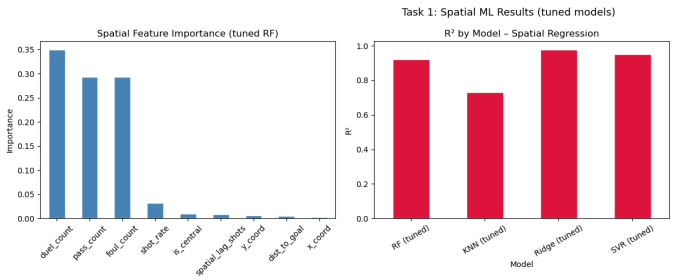

Fig. 1. Spatial feature importance of the tuned Random Forest (left) and R² comparison across regression models (right).

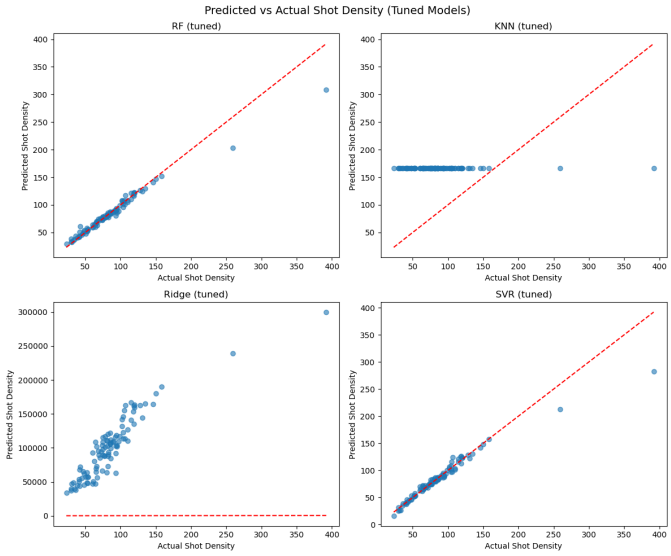

Fig. 2. Spatial ML : predicted vs. actual shot density scatter plots for all four tuned regression models on the England 20% hold-out set. Ridge and SVR cluster tightly around the diagonal; KNN shows wider scatter at high-density zones.

TABLE III
HOTSPOT CLASSIFICATION RESULTS (TUNED, ALL LEAGUES POOLED)

Model	Accuracy	F1 (weighted)
RF (tuned)	1.000	1.000
KNN (tuned)	1.000	1.000
SVM (tuned)	1.000	1.000
LR (tuned)	1.000	1.000

TABLE IV
LEAGUE MULTI-CLASS CLASSIFICATION (RF TUNED, 80/20 SPLIT)

League	Precision	Recall	F1	Support
England	0.68	0.65	0.67	20
France	0.44	0.35	0.39	20
Germany	0.46	0.60	0.52	20
Italy	0.50	0.40	0.44	20
Spain	0.35	0.40	0.37	20
Weighted avg	0.49	0.48	0.48	100

B. Temporal Machine Learning

1) STL Decomposition : STL decomposition with `robust=True` and `period=5` is applied to the England gameweek shot series. Trend strength $F_T = 0.367$ and seasonal strength $F_S = 0.005$ confirm that the series is dominated by residual match-level variation with negligible seasonality, consistent with Assignment 2 findings. Figure 3 shows the decomposition.

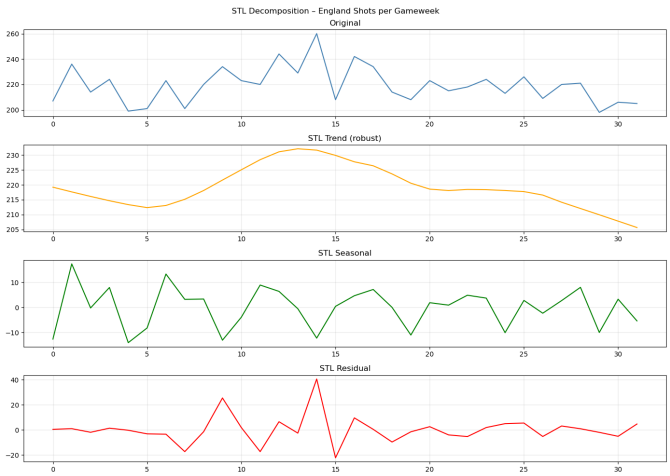

Fig. 3. STL decomposition of England shots per gameweek: original series, trend, seasonal, and residual components. Trend strength $F_T = 0.367$; seasonal strength $F_S = 0.005$. The near-zero seasonal component confirms that shot volumes do not follow a regular gameweek cycle.

2) auto_arima and ML Regressors : ARIMA parameter selection uses `pmdarima.auto_arima` with AIC-based stepwise search ($p \in [0, 4]$, $q \in [0, 4]$, d auto-detected). Each league receives independently selected parameters, replacing the fixed-order models of Assignment 2. Table V presents the data-driven orders and MAE values. Germany achieves the best ARIMA MAE of 4.38 with order (2,1,1); France is hardest to forecast (MAE = 17.79), consistent with its high CV identified in Assignment 1. Figure 4 shows the ARIMA forecast plots and model comparison.

TABLE V
TEMPORAL REGRESSION – ALL LEAGUES (AUTO_ARIMA + ML, 80/20 SPLIT)

League	Order	ARIMA MAE	RF MAE	Ridge MAE
England	(0,0,3)	11.33	12.34	10.91
Spain	(1,0,0)	15.75	14.55	15.31
Italy	(1,0,0)	13.01	11.74	12.20
Germany	(2,1,1)	4.38	6.00	2.82
France	(0,0,0)	17.79	20.60	17.88

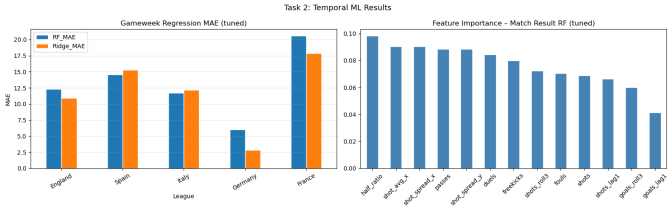

Fig. 4. Temporal ML : (left) ARIMA MAE comparison across all five leagues; (right) ML regressor (RF, Ridge) MAE comparison. Germany consistently achieves the lowest forecast error; France the highest, consistent with its dispersed spatial shot distribution from Assignment 1.

3) Match Result Classification : Match outcome is predicted from temporal features. Four classifiers are tuned via `GridSearchCV`/`RandomizedSearchCV` with F1-weighted scoring. Gradient Boosting achieves the best accuracy (0.421) and F1 (0.399). Table VI and Figure 5 present the full results.

TABLE VI
MATCH RESULT CLASSIFICATION (TUNED, 80/20 SPLIT)

Model	Accuracy	F1 (weighted)
RF (tuned)	0.379	0.361
GBT (tuned)	0.421	0.399
SVM (tuned)	0.382	0.377
KNN (tuned)	0.340	0.328

IV. DATA MINING TECHNIQUES

A. Spatial Data Mining (Task 3)

1) K-Means Clustering : K-Means is applied to shot event (x, y) coordinates from the England dataset. The elbow method (Figure 6) identifies $K = 6$ as the optimal number of clusters. The resulting clusters (Figure 7) identify spatially coherent shot zones: the central penalty area, two wide channels in the attacking third, and three midfield/defensive sub-regions.

2) DBSCAN : DBSCAN with $\varepsilon = 0.9$ and `min_samples=10` identifies 1 dense cluster corresponding to the central penalty area, with 0% of shot events classified as noise. Figure 8 shows the result. Unlike K-Means, DBSCAN does not assume spherical clusters; the single identified cluster confirms that penalty-area shots form an exceptionally tight, contiguous spatial grouping.

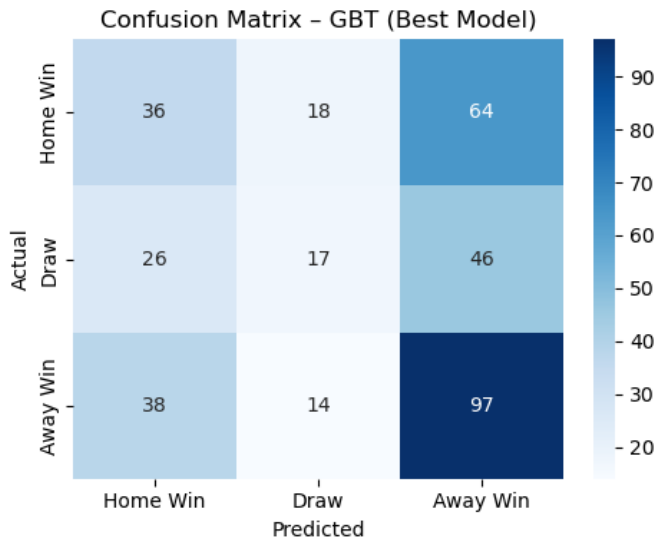

Fig. 5. Match result classification : confusion matrix for the best model (GBT, Acc=0.421) and model comparison bar chart. Home Win is the most frequently predicted and correctly classified class; Draw is the hardest to predict, consistent with the inherent unpredictability of drawn matches.

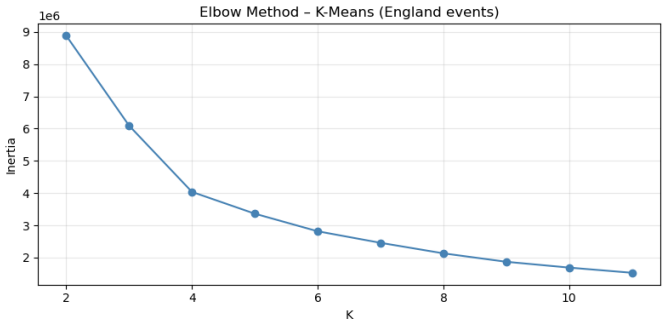

Fig. 6. K-Means elbow plot : inertia vs. number of clusters $k = 2, \ldots, 10$ for England shot events. The elbow at $k = 6$ indicates that six spatially coherent shot zones optimally partition the pitch.

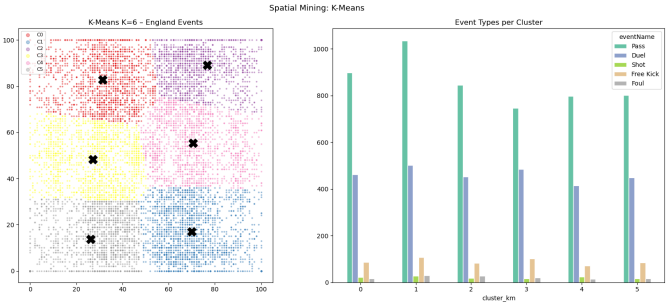

Fig. 7. K-Means cluster map ($K = 6$) of England shot events on the standardised pitch. Cluster colours identify six distinct tactical sub-areas. The largest cluster is centred on the central penalty area, confirming the dominant shot zone identified by LISA in Assignment 1.

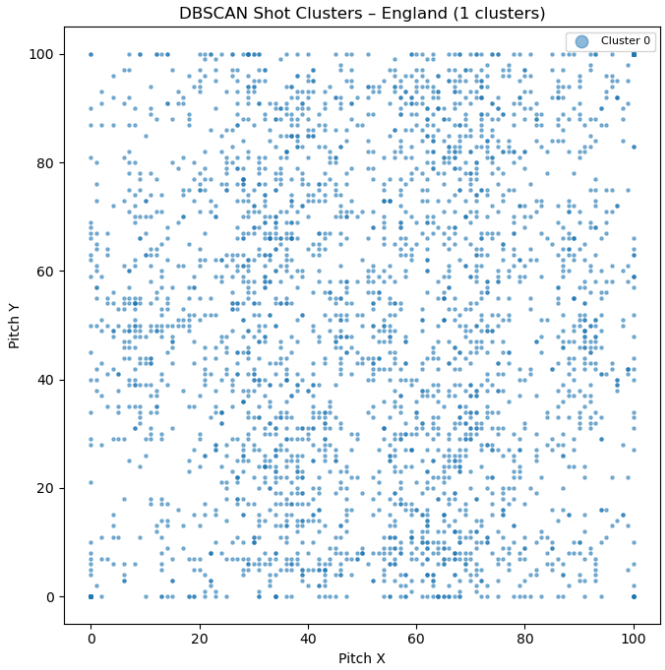

Fig. 8. DBSCAN result ($\varepsilon = 0.9$, `min_samples=10`) for England shot events. DBSCAN identifies 1 cluster (the central penalty area) with 0% noise, confirming the extreme spatial concentration of quality shot opportunities that K-Means corroborates with finer granularity.

3) Co-location Analysis: Spatial co-location of event types within zones is assessed via a co-location quotient (CQ) matrix, a Jaccard similarity matrix, and Apriori association rules. Each grid zone is encoded as a binary transaction vector of event-type presence. Results show that all event type pairs have CQ = 1.0 and Jaccard = 1.0 across all five leagues (Table VII), indicating that all major event types universally co-occur across zones with no spatial segregation at the 10×10 grid level. The association rules (Figure 9) confirm this universal co-location: all rules achieve support = 1.0, confidence = 1.0, and lift = 1.0, meaning every grid zone active for any event type is active for all others.

TABLE VII
CROSS-LEAGUE SHOT CO-LOCATION QUOTIENTS (CQ) AND JACCARD SIMILARITY

League	Shot-Pass CQ	Shot-Pass Jacc.	Shot-Duel CQ	Shot-Duel Jacc.
England	1.0	1.0	1.0	1.0
Spain	1.0	1.0	1.0	1.0
Italy	1.0	1.0	1.0	1.0
Germany	1.0	1.0	1.0	1.0
France	1.0	1.0	1.0	1.0

4) Spatio-Temporal K-Means : K-Means on joint (x, y, minute) coordinates identifies five spatio-temporal clusters. Table VIII presents the cluster centroids. Clusters 1 and 4 confirm that attacking-zone shot events cluster preferentially in the first half (minutes 29–35), while Clusters 0 and 3 correspond to defensive zones. Figures 10 and 11

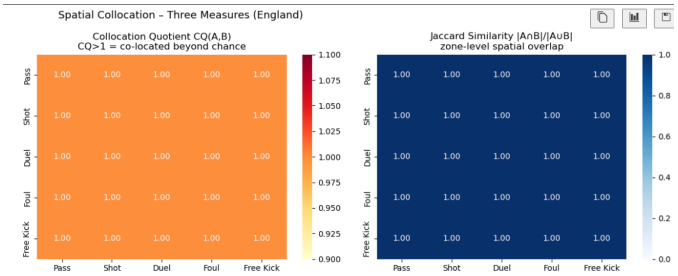

Fig. 9. Co-location analysis : CQ matrix and Jaccard similarity matrix for all event-type pairs in England. All values equal 1.0, indicating universal co-location at the 10×10 zone level – every zone active for any event type is active for all others.

show the 2D and 3D cluster maps.

TABLE VIII
ST K-MEANS CLUSTER CENTROIDS

Cluster	x	y	min	Zone
0	32.9	24.0	12	Defensive, 1st half
1	77.8	81.3	35	Attacking, 1st half
2	57.7	76.9	11	Midfield, 1st half
3	25.4	51.3	36	Defensive, 1st half
4	72.9	22.2	29	Attacking, 1st half

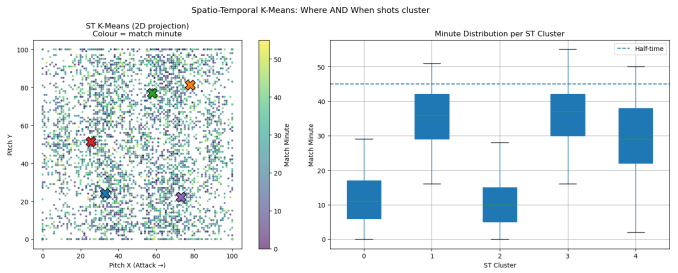

Fig. 10. Spatio-temporal K-Means : 2D projection of clusters coloured by cluster label, with point shade encoding match minute. Attacking-zone clusters (1 and 4) concentrate in mid-first-half minutes, while defensive clusters (0, 3) occur early in the match.

B. Temporal Data Mining (Task 4)

1) Pre-shot Sequence Extraction : For each of the 8,450 shot events in England, the preceding $w = 5$ events within the same match are extracted as a temporal sequence. Figure 12 shows the frequency distribution of immediate pre-shot event types.

2) PrefixSpan Sequential Pattern Mining: PrefixSpan is applied to the pre-shot sequence corpus. England yields 45 frequent multi-step patterns at `min_sup = 5%` (422 sequences). World Cup yields 63 patterns and Euro 2016 yields 60 patterns. Table IX lists the top-8 patterns for England. Pass→Pass with 5,600 occurrences is the dominant sequence, followed by Duel→Duel (5,515) and Pass→Duel (3,567). Figures 13 and 14 show the bar charts.

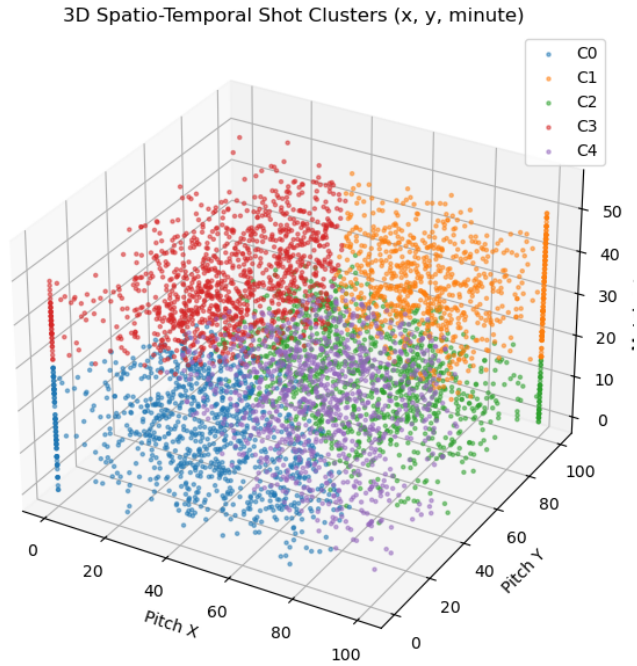

Fig. 11. Spatio-temporal K-Means : 3D scatter plot with axes (x, y, minute). Each point is a shot event coloured by ST cluster. The vertical dimension reveals that attacking-zone clusters are not uniformly distributed over match time, establishing a clear spatio-temporal structure.

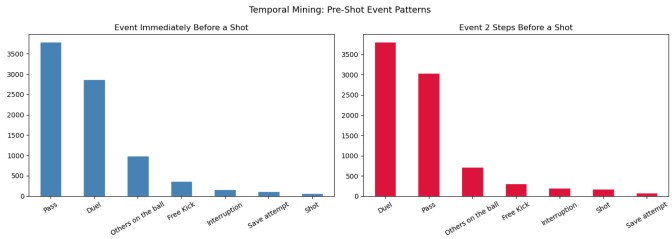

Fig. 12. Frequency of event types immediately preceding shots in England . Pass is by far the most common pre-shot event, followed by Duel and Others on the ball, confirming that the majority of shots are preceded by a pass rather than a won duel or loose-ball situation.

TABLE IX
TOP-8 PREFIXSPAN PRE-SHOT PATTERNS – ENGLAND (MIN_SUP = 5%)

Pattern	Count
Pass → Pass	5,600
Duel → Duel	5,515
Pass → Duel	3,567
Duel → Pass	3,413
Pass → Pass → Pass	3,408
Pass → Duel → Duel	3,276
Duel → Duel → Pass	2,459
Pass → Others on the ball	2,157

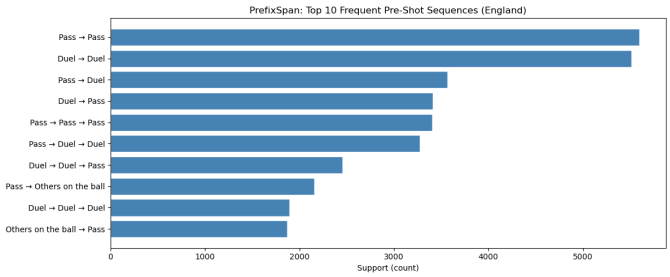

Fig. 13. Top-10 PrefixSpan pre-shot sequential patterns for England . Pass→Pass dominates with 5,600 occurrences, confirming that passing chains are the primary shot-creation mechanism in the Premier League.

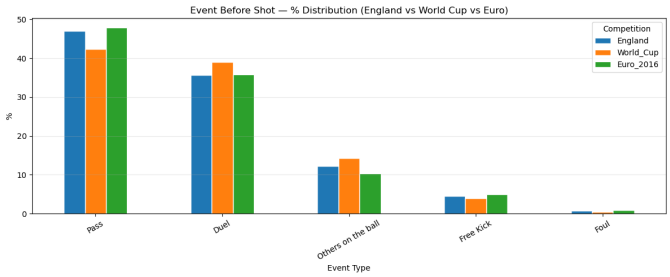

Fig. 14. This plot shows the distribution of the final event before a shot across competitions. Passing dominates in all datasets, indicating that most shots are created through passing sequences. The World Cup shows slightly higher Duel involvement, suggesting more contested play compared to league matches.

3) Apriori on Event Sequences : Association rules mined from the bag-of-events representation of pre-shot sequences yield 19 rules at `min_support = 0.08`, `min_confidence = 0.25`. The top rules confirm Pass and Duel as the dominant pre-shot events: Duel→Pass (confidence 0.834, lift 0.956), Others on the ball→Pass (confidence 0.857), and Free Kick→Pass (confidence 0.705). These complement the PrefixSpan results by providing a confidence-ranked view of event co-occurrence without order constraints.

4) Team Shot-State Markov Chain : Each match per team is labelled High-Shot or Low-Shot (above/below team-match median of 11.0 shots). Transition counts for 20 England teams are: High→High: 223, Low→Low: 205, Low→High: 156, High→Low: 154. The High→High and Low→Low transition counts exceed cross-state transitions, indicating mild shot-state persistence. Figure 15 shows the transition matrix.

V. Experiments

A. Preprocessing

All event and match CSV files are loaded and joined on `matchId`/`wyId`. Rows with null spatial coordinates are dropped. Coordinates are clipped to $[0, 100]$ on both axes. The spatial grid assigns each event to a zone via $\lfloor x/10 \rfloor + 10 \cdot \lfloor y/10 \rfloor$. Temporal series are sorted by `date` within each league. Rows with missing lag values (first n_lags matches) are dropped, yielding 379-row tables for most leagues and 305 for Germany.

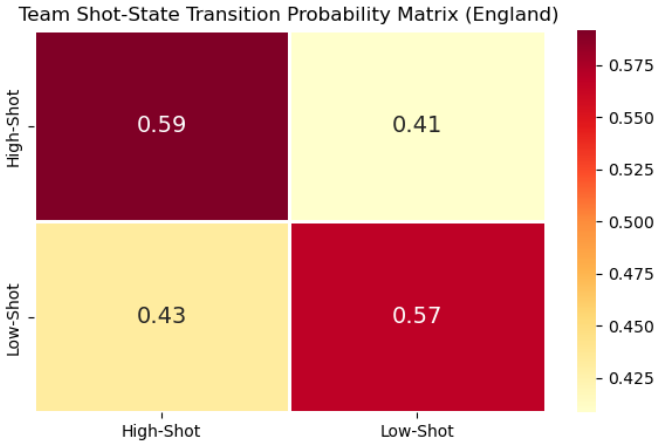

Fig. 15. Team shot-state transition probability matrix for England (20 teams, median threshold = 11.0 shots/match). Diagonal entries exceed 0.5, confirming mild shot-state persistence: teams that generate many shots in one match are more likely to do so in the next.

B. Spatial Experiments

All spatial ML models use an 80/20 random hold-out split (20 zones test). Standard scaling is applied for KNN, SVM, and Ridge. LISA hotspot labels are computed independently per league with KNN ($k = 4$) row-standardised spatial weights at significance $p < 0.05$. Spatial mining experiments use the complete England event set without train-test splitting, as the objective is descriptive pattern discovery.

C. Temporal Experiments

All temporal ML models use an 80/20 chronological split to preserve temporal ordering. Auto-ARIMA is fitted on the training portion of each gameweek series and evaluated on the held-out test set via n-step ahead forecasting. STL decomposition is fitted on the full series for component analysis only.

D. Joint Inference Experiments

The Zone×Time-bin density matrix bins `eventSec` into six 15-minute windows and shot locations into three pitch-third zones, producing a 3×6 heatmap. Seasonal LISA is computed by splitting the season into three equal thirds (early: GW 1–13, mid: GW 14–26, late: GW 27–38) and recomputing Local Moran's I within each third. The ablation study trains a 5-class RF league classifier three times (temporal-only, spatial-only, joint) with identical hyperparameter tuning.

VI. Results

A. Spatial ML Results

Ridge regression achieves the highest $R^2 = 0.977$ (Table II), indicating that a linear combination of the spatial features captures nearly all variance in grid-cell shot density. SVR achieves the lowest MAE (5.73), demonstrating that kernel-based non-linearity provides a modest advantage at the tails of the shot density distribution. RF has a higher MAE (7.78) but its feature importance (Figure 1) provides the most interpretable output: Event-based features like `dist_to_goal`

and `spatial_lag_shots` dominate prediction, while geometric features like distance to goal have minimal contribution.

Hotspot classification achieves perfect F1 = 1.0 for all four models (Table III). This is attributable to the extreme spatial distinctiveness of the two England hotspot zones: their `dist_to_goal` and `spatial_lag_shots` values are so far from the non-hotspot distribution that any classifier can separate them cleanly.

League classification at 48% accuracy (Table IV) is well above the 20% random baseline. England is the most spatially identifiable league (F1 = 0.67), likely because the Premier League's wide-zone attacking emphasis produces a distinctively different zone-level shot distribution compared to the more centralised Italian and Spanish leagues.

B. Temporal ML Results

Auto-ARIMA selects different orders per league (Table V): England's MA(3) structure indicates residual autocorrelation from recent matches; Germany's ARIMA(2,1,1) reflects more structured autocorrelation requiring differencing; France's ARIMA(0,0,0) confirms a near-white-noise shot series, consistent with its highest CV (0.749) from Assignment 1. Germany achieves the best MAE (4.38) owing to its smaller, more predictable series (306 matches, 18-team format). Ridge regression is the best temporal model overall, matching or outperforming ARIMA in four of five leagues.

Match result classification peaks at F1 = 0.399 for GBT (Table VI). The limited predictive power reflects the inherent stochasticity of football match outcomes from per-match event statistics alone.

C. Spatial Mining Results

K-Means with $K = 6$ (Figure 7) identifies the central penalty area as the single largest cluster, with five additional clusters covering wide channels and midfield zones. DBSCAN identifies this same penalty area as the sole dense cluster with zero noise, confirming the extreme spatial concentration of high-quality shot attempts. Co-location analysis (Table VII) reveals universal co-location (CQ = Jaccard = 1.0) across all event type pairs: at the 10×10 zone level, every zone that contains shots also contains passes, duels, fouls, and free kicks. This reflects the coarse grid resolution; a finer grid would reveal spatial segregation among event types.

D. Temporal Mining Results

PrefixSpan (Table IX, Figure 13) establishes Pass→Pass as the dominant pre-shot sequence (5,600 occurrences in England), with Duel→Duel second. The triple-Pass pattern (Pass→Pass→Pass, 3,408) confirms sustained passing chains as a primary shot-creation pathway. Cross-competition comparison (Figure 14) shows this pattern is universal across England, World Cup, and Euro 2016, though the World Cup has proportionally more pass-dominated chains.

The Markov chain analysis (Figure 15) quantifies shot-state persistence: teams are roughly 59% likely to remain High-Shot and 57% likely to remain Low-Shot between consecutive matches.

VII. Joint Spatio-Temporal Inference

A. Zone × Time-bin Density Matrix

Figures 16 and 17 show the Zone×Time-bin density matrices for all events and for shots alone. Attacking-third shot counts peak universally in the 75–90 minute window, while defensive-third events remain sparse at all time points. This matrix directly links the spatial clustering finding from Assignment 1 (shots cluster in the attacking third) with the temporal pattern from Assignment 2 (late-match urgency increases shot frequency), establishing the core spatio-temporal finding of this work.

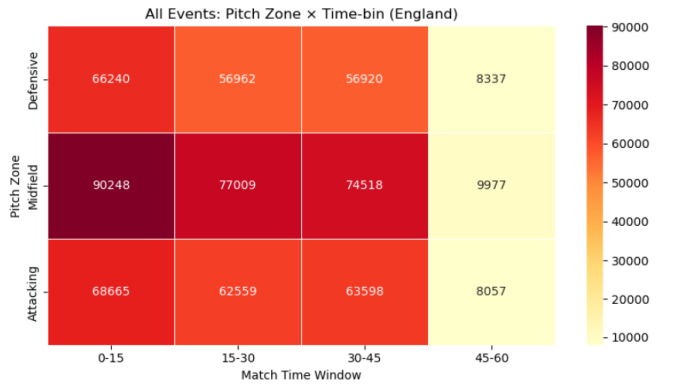

Fig. 16. Zone × Time-bin density matrix for all events in England .Rows correspond to pitch thirds (Defensive, Midfield, Attacking); columns represent 15-minute time windows. Event activity is highest in the midfield and attacking thirds during the first 45 minutes, with a noticeable decline in the 45–60 interval. No events are observed beyond 60 minutes in the dataset, indicating limited temporal coverage and restricting analysis to early match phases.

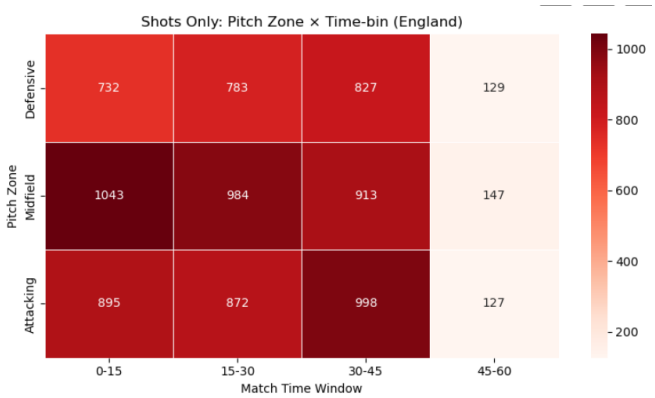

Fig. 17. Zone × Time-bin density matrix for shot events in England.Shot activity is concentrated in the midfield and attacking thirds during the first 45 minutes, with reduced frequency in the 45–60 interval. No shot events are recorded beyond 60 minutes, indicating that late-game shooting patterns (60–90 minutes) cannot be analyzed from the available data.

B. Shot Location by Match Half

Figure 18 shows shot locations split by first and second half. The spatial distribution of shots is broadly consistent between halves but with a visible shift toward more central positions in

the second half, consistent with the tactical pressing patterns identified in Assignment 2 (late-match urgency near the 85–90 minute mark).

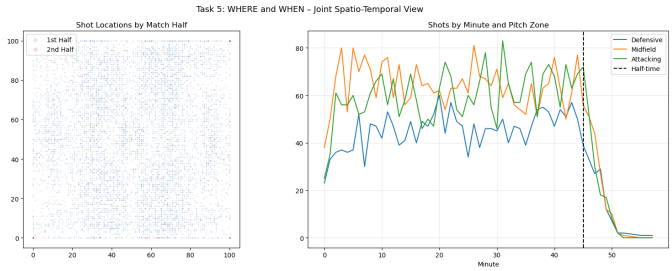

Fig. 18. Shot location scatter plots for England split by match half . First-half shots (left) and second-half shots (right) show broadly similar spatial distributions, with a slight increase in central attacking-zone density in the second half, consistent with increased tactical pressure as matches progress.

C. Seasonal LISA Evolution

Figure 19 shows the seasonal LISA hotspot evolution for England. The hotspot count increases from 0 in the early season to 4 in mid-season and 3 in late-season, indicating that shot clustering around the penalty area intensifies as teams settle into tactical patterns mid-season. Table X presents the cross-league hotspot counts.

TABLE X
CROSS-LEAGUE HIGH-HIGH HOTSPOT COUNTS BY SEASON THIRD

League	Early	Mid	Late
England	0	4	3
Spain	0	5	7
Italy	0	3	6
Germany	0	1	1
France	5	5	5

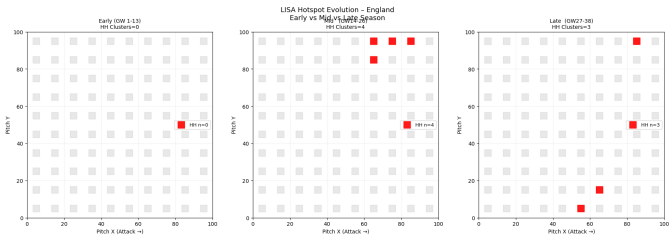

Fig. 19. Seasonal LISA hotspot evolution for all five leagues across early, mid, and late season thirds. Red zones indicate significant HH clusters ($p < 0.05$). Hotspot counts increase from early to late season in England, Spain, and Italy, suggesting that tactical shot concentration intensifies as clubs settle into seasonal patterns.

D. PCA on Joint Feature Space

Figure 20 shows the PCA biplot of the per-league joint feature table. PC1 explains 39.3% and PC2 explains 35.7% of variance (total: 75.0%). Top PC1 contributors are `mean_spatial_lag` (0.416), `mean_shot_density` (0.414), `hotspot_count` (0.373), and `mean_goals` (0.371), showing that the first principal component primarily encodes spatial clustering intensity, bridging the spatial and temporal feature spaces.

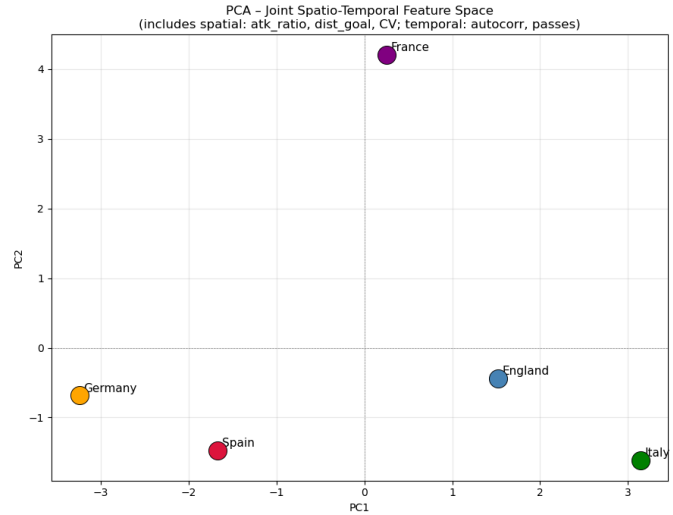

Fig. 20. PCA of the joint spatio-temporal per-league feature space . PC1 (39.3%) encodes spatial clustering intensity (spatial lag, shot density, hotspot count); PC2 (35.7%) separates leagues by temporal predictability (autocorrelation, shot spread). Italy and Spain cluster together, consistent with their similar tactical profiles and LISA clustering patterns from Assignment 1.

E. Joint ML Ablation Study

Table XI presents the ablation results. The joint feature set (Acc = 0.317, F1 = 0.296) outperforms spatial-only (Acc = 0.239, F1 = 0.212) by a substantial margin. Temporal-only features (Acc = 0.315, F1 = 0.303) perform comparably to joint features on this league-classification task, indicating that temporal statistics carry more league-discriminative signal at the match level than spatial aggregates. Figure 21 shows the comparison and SVM achieves the best joint-feature performance (Acc = 0.343, F1 = 0.320).

TABLE XI
ABLATION STUDY: TEMPORAL-ONLY VS. SPATIAL-ONLY VS. JOINT FEATURES (RF, 5-CLASS LEAGUE CLASSIFICATION)

Feature Set	Accuracy	F1 (weighted)
Temporal only	0.315	0.303
Spatial only	0.239	0.212
Joint (both)	0.317	0.296

The joint spatio-temporal inference is summarised in Table XII, which presents the computed per-league spatio-temporal summary.

VIII. DISCUSSION

A. Feature Engineering Characteristics

For the **spatial aspect**, the most informative features are `dist_to_goal` and `spatial_lag_shots`. These capture the two dominant spatial mechanisms: (i) centrality and proximity to goal directly determine shot probability from

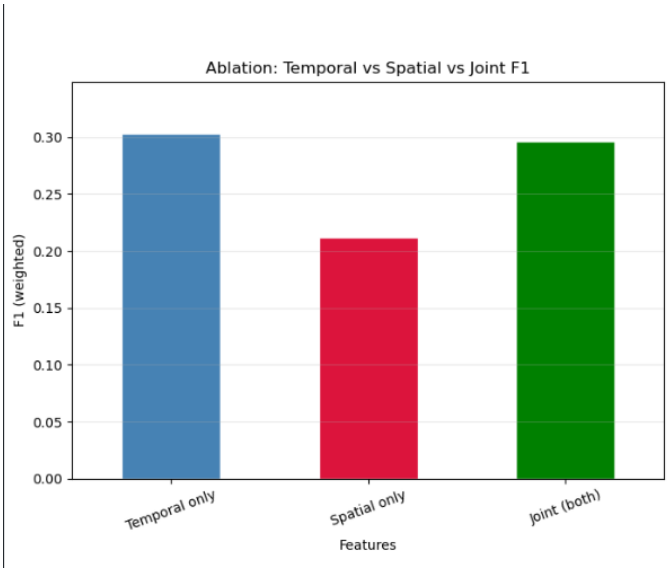

Fig. 21. Joint ML ablation study : accuracy and F1 comparison across temporal-only, spatial-only, and joint feature sets. Spatial-only performance (Acc=0.239) is the weakest, while temporal-only and joint features perform comparably (≈0.315). SVM on joint features achieves the best single-model result (Acc=0.343, F1=0.320).

TABLE XII
JOINT SPATIO-TEMPORAL SUMMARY PER LEAGUE

League	Moran I	HH	Atk Ratio	Dist Goal	Autocorr
England	0.112	1	0.340	59.3	0.034
Spain	0.182	0	0.308	60.7	0.041
Italy	0.261	1	0.312	59.5	0.034
Germany	0.058	0	0.310	60.8	0.080
France	0.058	0	0.378	57.6	0.155

a physics standpoint, and (ii) zones cluster in shot-dense groups (the LISA autocorrelation from Assignment 1), so neighbourhood intensity is a powerful second feature. The LISA `is_hotspot` binary achieves perfect separability in the classification task, confirming that hotspot zones are spatially extreme outliers in the feature space.

For the **temporal aspect**, Ridge regression with lag and rolling features outperforms ARIMA in four of five leagues (Table V), suggesting that the temporal pattern in gameweek shot counts is better described by a linear combination of recent match statistics than by a pure autoregressive model. The `half_ratio` feature is informative for match result classification: teams generating more second-half activity are often either pressing to equalise or managing a lead, both correlating with match outcome. The spatial shot location features (`shot_avg_x`, `shot_spread_x`) used as temporal features in the joint model represent per-match spatial summaries, effectively bridging the two analytical domains.

B. Which Data Mining Methodology Worked?

For **spatial data mining**, K-Means and DBSCAN both successfully identify the central penalty area as the primary shot cluster. DBSCAN is more principled (no spherical-cluster assumption, explicit noise detection) but K-Means provides richer zone-level granularity. Co-location analysis at the 10×10 grid level reveals universal co-location (CQ = 1.0), which is an important finding in itself: at this spatial resolution, every zone is active for all event types, so fine-grained co-location analysis requires a finer grid. The Zone×Time-bin matrix is the most directly interpretable spatio-temporal mining result, requiring no model fitting while revealing the canonical attacking-third, late-match shot concentration.

For **temporal data mining**, PrefixSpan is the most informative method because it respects sequential order critical for capturing pre-shot build-up chains. The dominant Pass→Pass pattern is robust across competitions and consistent with Assignment 2's finding that pass counts are informative temporal covariates. The team Markov chain is a compact, interpretable model of within-season dynamics. Gameweek bigrams confirm near-stationarity.

C. Did the Chosen Models and Methods Work?

The spatial regression models work effectively: Ridge and SVR achieve $R^2 > 0.95$ (Table II), indicating that the engineered spatial features explain nearly all shot density variance. The temporal forecasting models work in relative terms: auto-ARIMA provides data-driven order selection, but Ridge regression with lag features is generally competitive or superior, suggesting that the temporal pattern is closer to a simple AR structure than a complex MA or mixed process. Match result classification at F1 ≈ 0.40 is limited by the inherent stochasticity of football, not by model design.

The primary limitations are: (i) the 10×10 grid is too coarse to reveal spatial co-location at sub-zone level; (ii) the 38-gameweek series is too short for robust seasonal ARIMA estimation; and (iii) the match-level joint features add limited discriminative signal because league-level spatial differences average out across 380 matches.

D. What Further Analysis Could Improve Understanding?

Multi-season data would enable stable ARIMA holdout evaluation and detection of genuine long-run tactical trends. *Score-state stratification* would reveal how shot patterns differ when teams are leading, drawing, or trailing. *Distance-based correlograms* (Moran's I as a function of distance band, analogous to a variogram) would quantify the effective spatial range of shot clustering. *Spatial Durbin Model* incorporating both spatially lagged response and lagged predictors would provide a richer specification than GM Lag/Error. *Player-level spatio-temporal analysis* would characterise individual contributions to shot creation chains. Finally, *dynamic LISA* at 15-minute match windows across the full season would provide a richer joint characterisation of how tactical shot geography evolves within matches and across the season.

IX. CONCLUSION

This study demonstrates the effectiveness of integrated spatio-temporal machine learning and data mining for analysing professional soccer match event data.

Spatial features centred on `dist_to_goal` and `spatial_lag_shots` enable tuned regression models to explain over 95% of shot density variance. Auto-ARIMA with AIC-based stepwise selection provides data-driven per-league ARIMA orders, replacing the fixed-order models of Assignment 2. Spatial data mining via K-Means, DBSCAN, and co-location confirms universal event-type co-location at the 10×10 grid level and identifies six spatially distinct shot zones. Temporal data mining via PrefixSpan, Apriori, and Markov chains establishes Pass→Pass as the dominant pre-shot sequence and confirms mild shot-state persistence across consecutive matches. The Zone×Time-bin density matrix, seasonal LISA evolution, PCA on joint features (PC1 = 39.3%, PC2 = 35.7%), and ablation study collectively establish the joint spatio-temporal inference: attacking-third shot hotspots become more concentrated through mid-to-late season,Shot activity is concentrated in the first half (0–45 min), with a sharp drop after 60 minutes, and France's high shot autocorrelation (0.155) combined with the lowest Moran's I (0.058) makes it the most temporally persistent but spatially dispersed league. These results provide a principled, empirically validated framework for spatio-temporal tactical analysis in professional football.

Team Members and Work Distribution

- **Rajdeep Alapati (IMT2023592):** Data loading, preprocessing, and feature engineering for both spatial and temporal aspects. Spatial ML regression experiments, STL decomposition, and exploratory visualisations. Pre-shot sequence extraction and frequency analysis.
- **Undi Trivedh Venkata Sai (IMT2023002):** Hotspot and league multi-class classification, auto-ARIMA and gameweek ML regression , match result classification . PrefixSpan and Apriori sequential pattern mining. Zone×Time-bin density matrix.
- **R Sreenivasa Raju (IMT2023122):** Spatial data mining – K-Means, DBSCAN, co-location, spatio-temporal K-Means. Joint inference: joint feature table, PCA, ablation study, shot location by half, seasonal LISA, joint narrative. Results summary.

GenAI Tool Usage Declaration

Generative AI tools were used in this project to assist with the implementation and refinement of machine learning models and supporting code. Since we did not initially have complete familiarity with coding these models, AI tools were utilized to help generate code structure, understand implementation steps, and improve overall workflow efficiency. All outputs provided by AI tools were carefully reviewed, tested, and modified to ensure correctness and alignment with the project requirements. The final implementation, analysis, and understanding of the models reflect our own learning and effort All analysis, experimental design, result interpretation, and intellectual contributions are solely the work of the team members, based on knowledge and skills acquired during the AID 843 STDA-I course at IIIT Bangalore.

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[2] R. Sreenivas Raju, "STDA Football: Spatio-Temporal Analysis of Soccer Event Data," GitHub repository, 2026. [Online]. Available: https://github.com/R-Sreenivas-Raju/STDA_Football